SmartRoute Pro — Cloud Computing & DevOps Report

Subject: Cloud Computing (CC) • 3-Tier Architecture, Containerization, Health Probes & CI/CD

1. Three-Tier Cloud Topology & Service Model

SmartRoute Pro follows a Platform-as-a-Service (PaaS) deployment paradigm across three discrete tiers:
• **Presentation Tier:** React 19 + TypeScript 5.8 SPA bundled via Vite, hosted on Render Static Site or containerized Nginx.
• **Application Tier:** Python 3.11 Flask API running Gunicorn WSGI with the gthread worker model. ADA and SDA algorithm engines are executed directly within this process to avoid distributed network overhead.
• **Persistence Tier:** Managed PostgreSQL 16 DBaaS utilizing SQLAlchemy ORM and version-controlled Alembic migrations.

2. Container Specifications & Non-Root Security

Component	Container Specification	Security & Health Probes
Backend WSGI	python:3.11-slim, Gunicorn gthread	USER appuser (UID 1001), curl /api/health
Frontend Web	Stage 1: node:20-alpine (build) Stage 2: nginx:1.25-alpine	SPA fallback (try_files $uri /index.html), wget spider
Managed Database	postgres:16-alpine (Local Compose) Render Managed PostgreSQL (Cloud)	pg_isready -U postgres, persistent volume pgdata

3. Managed PostgreSQL Connection Pool Governance

Parameter	Configured Value	Architectural Rationale
pool_size	10 connections	Maintains warm TCP connections to reduce TLS handshake overhead
max_overflow	20 connections	Accommodates burst traffic spikes during batch routing queries
pool_recycle	1800 seconds (30m)	Recycles idle connections to avoid cloud router timeouts
pool_pre_ping	True (Active)	Issues test ping (SELECT 1) before handing connections to queries
Production Enforcement	Fail-Closed (RuntimeError)	**Strictly prohibits SQLite fallback in production environments**

4. Continuous Integration (CI/CD) & Security

GitHub Actions Pipeline (.github/workflows/ci.yml):
• Stage 1: Checkout & Runtime Environment Setup (Python 3.11 & Node 20 with dependency caching).
• Stage 2: Pytest Suite Automation (verifying all 55+ test cases across ADA, SDA, CRUD & Cloud Probes).
• Stage 3: Frontend TypeScript Typecheck (tsc -b) & Production Bundling (vite build).
• Stage 4: Docker Configuration Linting (validating docker-compose.yml syntax).
Security Controls: Response headers (X-Content-Type-Options: nosniff, X-Frame-Options: DENY, X-XSS-Protection), request payload ceiling (16MB), zero secret leakage in monitoring APIs, and optional mutation authorization token check.